Last time If f is cts. and injective, then it is (strictly) monotone.

Converse If f is strictly monotone, then f is injective, then f^{-1} exists.

Pf. If f is strictly monotone, for any $a, b \in D_f$ with $a \neq b$

$f(a) < f(b)$ so $f(a) \neq f(b)$

i.e. f is injective.

∎

Rmk If $f: I \to \mathbb{R}$ cts. & injective

- Image of f is also an interval.

Strategy If $a, b \in$ Image(f), then WTS $\forall c \in (a, b)$

$c \in$ Image(f) Using IVT

Pf Since $a, b \in$ Image(f)

$\exists x, y \in I$ s.t. $f(x) = a$, $f(y) = b$

Want $z \in D_f$ s.t. $f(z) = c$ $\forall a < c < b$

Apply IVT to $f(x) - c$ on (x, y) b/c. $f(x) - c < 0 < f(y) - c$

- If f is injective & cts., then f takes open intervals to open intervals.

Ex. • $f(x) = c$ is not injective, takes any open interval to $\{c\} = [c, c]$

- $f(x) = x^2$, takes $(-1, 1)$ to $[0, 1)$.

Thm If f is cts. & injective on I, then f^{-1} is also cts.

Pf By the previous theorem, f is strictly monotone

WoLoG say f is strictly increasing.

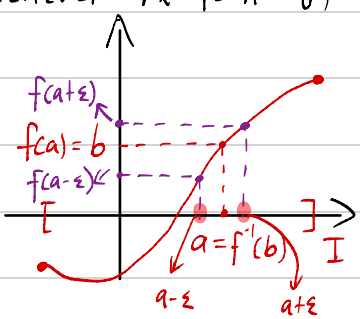
Want that f^{-1} is cts.

i.e. Want ~~$\lim_{x\to b} f^{-1}(x) = f^{-1}(b)$~~ for any $b \in D_{f^{-1}}$)

$b \in D_{f^{-1}}$ means that $\exists a$ s.t. $f(a) = b$

$\iff \lim_{x\to f(a)} f^{-1}(x) = a$

Let $\varepsilon > 0$. want to find $\delta > 0$ s.t. whenever $|x - f(a)| < \delta$, we have $|f^{-1}(x) - a| < \varepsilon$.



From injectivity & f strictly increasing

We know $(f(a-\varepsilon), f(a+\varepsilon))$ maps to $(a-\varepsilon, a+\varepsilon)$ via f^{-1}.

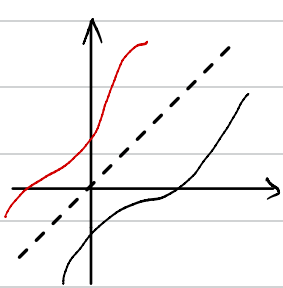
Let $\delta = \frac{1}{2}\min\{f(a) - f(a-\varepsilon), f(a+\varepsilon) - f(a)\}$, and by construction, this δ works for the given ε.

∎

Q What is $(f^{-1}(x))'$?

$f'(a) =$ Slope of tangent

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$



Thm If f is cts. + injective on an interval & $f(a)=b$
If $f'(a)=0$ then f^{-1} is not differentiable at b

Pf Suppose for a contradiction that it were differentiable
By the definition of the inverse function, $f(f^{-1}(x)) = x$ (¥)
Since f, f^{-1} are assumed to be differentiable, differentiate (¥)
$f'(f^{-1}(x)) \cdot (f^{-1}(x))' = 1$
If $f^{-1}(b)=a$, then $\underbrace{f'(a)}_{=0} \cdot (f^{-1}(x))'\Big|_{x=b} = 1$

This is a contradiction
$\Rightarrow f^{-1}$ not diff. at b



Thm Let f be cts. + injective, & Suppose f diff. at a w/ $f'(a)\neq 0$
Then f^{-1} is diff. at $b=f(a)$ & $(f^{-1})'(b) = \dfrac{1}{f'(f^{-1}(a))}$.

pf Want the limit

$$\lim_{h\to 0} \frac{f^{-1}(b+h)-f^{-1}(b)}{h} \quad \text{exists \&} = \frac{1}{f'(f^{-1}(a))}$$

$$= \lim_{h\to 0} \frac{f^{-1}(b+h)-a}{h}$$

Since f is inj + cts. the interval $(b-h, b+h)$ corresponds to $(a-k(h), a+k(h))$

$$= \lim_{h\to 0} \frac{f^{-1}(f(a)+k(h))-a}{f(a+k(h))-b}$$

$$= \lim_{h\to 0} \frac{a+k(h)-a}{f(a+k(h))-f(a)} = \lim_{k\to 0} \frac{k}{f(a+k)-f(a)}$$

$$= \frac{1}{f'(a)}$$